A Hybrid Simulation and Machine Learning Framework for Intrusion Detection in Vehicular Ad-Hoc Networks (VANETs)

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***Abstract*—Vehicular Ad-Hoc Networks (VANETs) form a foundational pillar of future Intelligent Transportation Systems (ITS) by enabling vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) data exchange for road safety, traffic optimization, and autonomous coordination. However, the dynamic and decentralized nature of VANETs introduces vulnerabilities that conventional cryptographic protection cannot fully mitigate. This paper presents a hybrid framework that couples the simulation of ns-3 networks with supervised machine learning (ML) classifiers for the detection of malicious nodes in VANETs. Using a labeled dataset combining simulated mobility, communication, and environmental features, we evaluate Support Vector Machine (SVM), Random Forest (RF), Logistic Regression (LR), K-Nearest Neighbors (KNN), and Multi-Layer Perceptron (MLP) models. Experiments show that nonlinear models outperform linear ones; the MLP achieves 82% accuracy and strongest F1-score, with latency, packet loss, and RSSI variance emerging as key predictive indicators. The results highlight the feasibility of integrating ML-based intrusion detection into real-time VANET simulations and open directions for 5G-V2X-secured ITS infrastructures.**

***Index Terms*—VANET, Intrusion Detection, ns-3, Machine Learning, Sybil Attack, DoS, Spoofing, Intelligent Transportation Systems**

I. Introduction

Connected vehicles form a core component of next-generation mobility. Through VANETs, vehicles broadcast and receive periodic beacons (e.g., Basic Safety Messages, BSMs) to maintain cooperative awareness. These messages include speed, position, and heading information, enabling applications such as collision avoidance and lane merging.

However, the same openness that allows interoperability also enables adversaries to infiltrate. Attackers can impersonate other nodes (Spoofing), generate multiple fake identities (Sybil), or flood the network with bogus messages (DoS). Such behavior can degrade channel capacity, mislead routing decisions, and compromise driver safety.

Traditional security frameworks rely on Public Key Infrastructure (PKI) or pre-shared keys; yet, these approaches cannot handle dynamic vehicle arrivals, key revocation delays, or insider threats. Hence, data-driven anomaly detection that adapts to context—such as ML-based IDS—is critical.

This paper focuses on detecting malicious network behavior using supervised learning models trained on synthetic and simulation-derived data. The hybrid design ensures both realism (through simulation ns-3) and scalability (through data-driven modeling).

II. Related Work

Early detection of VANET intrusion relied on rule-based systems or cryptographic verification [2]. However, attackers can exploit mobility and propagation variability to bypass static thresholds.

Machine learning approaches have since emerged. Decision Trees and SVMs have been used for Sybil detection [1], while ensemble methods improve detection accuracy under noise. Neural networks further enhance generalization to unseen attack behaviors, but require large and diverse datasets.

Our framework differs by directly integrating ns-3-generated communication traces—incorporating realistic radio channel fluctuations, packet collisions, and vehicle mobility—into the ML training pipeline. This hybridization bridges the gap between simulation-based performance study and real-world IDS training.

We have based our project on various different technical papers, which are as follows.

A. SVM-based Detection of False Data Injection in Intelligent Transportation System [3]

This paper presents a machine learning-based Intrusion Detection System (IDS) for VANETs to detect False Data Injection attacks. Using a MATLAB/NS-3 co-simulation framework, deploying SVM classifiers in each vehicle to classify received packets as trusted or malicious.

The IDS was tested under various conditions including different vehicle densities on the road, attacker numbers, and road conditions such as with and without congestion. The system's performance was evaluated using the classification rate, attack detection rate, false positive rate, and detection speed metrics. Finally, the results demonstrated high precision in differentiating between malicious data from trusted data, successful reduction of average vehicle distance errors after attack mitigation, and suitable detection times for truck platoon applications.

B. A Comparative Study on Communication Architecture of Unmanned Aerial Vehicles and Security Analysis of False Data Dissemination Attacks [4]

This work addresses the challenges of path failures, high mobility, disconnected topology, and traffic density in VANET by proposing a UAV-supported architecture that provides V2U/U2V (VANET to UAV/ UAV to VANET) communication. Presents a decentralized multi-layered UAV ad-hoc network, FANET for improved VANET reliability by researching various possible communications architectures. To offer a comparative study on security vulnerabilities in UAV-supported vehicular networks, this paper investigates various possibilities of false data dissemination attacks and their impacts resulting from different architectural configurations.

This paper presents a network, supporting VANETs, which provides efficient networking with much better scalability and without single-point failure. Compares several UAV network architectures, highlighting limitations including the following: centralized architecture has a single point of failure; UAV ad-hoc networks have network overhead; multi-group UAV architecture is not robust. Analyzes the impact of false data dissemination attacks, which are identified as the most dangerous cyber-attack, on these architectures.

C. Machine Learning for Intrusion Detection in Vehicular Ad-hoc Networks (VANETs): A Survey [5]

It showcases a comprehensive survey on machine learning-based intrusion detection systems in VANET. The survey analyzes the latest ML techniques in VANET intrusion detection by examining how these systems monitor network traffic to identify suspicious activities and implement countermeasures necessary to protect the integrity and security of connected vehicles.

This survey has presented a comprehensive analysis of machine learning-based intrusion detection systems for VANETs and has identified that both known and unknown security threats can be detected by techniques such as anomaly detection, signature-based detection, and ML algorithms. The work serves as a valuable resource focused on comparative analysis with an aim to establish the optimal approaches and datasets for specific use cases, which will enable achieving the full potential of VANETs for improving safety and efficiency in traffic. The work concludes by highlighting current challenges and discussing future research directions in the area of VANET security.

The proposed framework integrates simulation, data preparation, ML training, and evaluation.

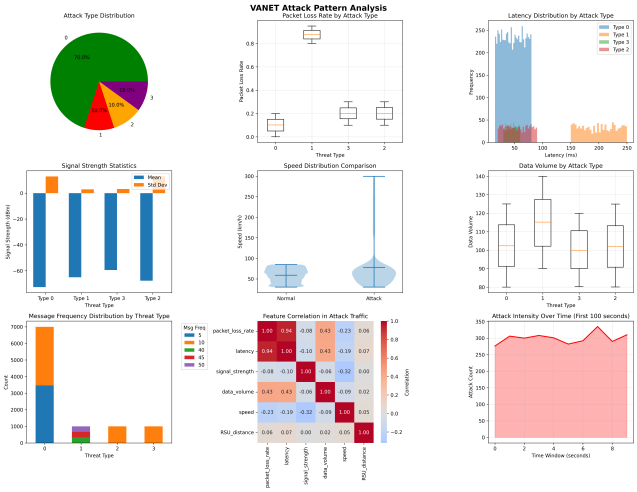

Fig. 1: Framework overview showing ns-3 VANET simulation, data extraction, machine learning training, and evaluation pipeline.

III. System Architecture

A. Network Simulation in ns-3

The simulation uses IEEE 802.11p PHY/MAC with a single Road-Side Unit (RSU) serving multiple mobile nodes. Vehicles follow a mobility trace defined by the RandomWaypoint model, achieving speeds between 20–35 m/s. Each node transmits periodic BSM packets at 10 Hz.

Three attack modules are implemented:

- **DoS**: A node floods the medium with high-frequency packets, raising latency and loss.
- **Spoofing**: A node clones another ID but maintains a mismatched RSSI and position vector.
- **Sybil**: A node spawns several pseudonymous identities broadcasting identical trajectories.

B. Dataset Composition

The dataset consists of 10,000 events with features spanning:

- *Mobility:* speed, acceleration, lane position, heading.
- *Communication:* RSSI, packet loss ratio, end-to-end latency.
- *Environment:* traffic density, road type, weather condition.

Data are normalized using z-score scaling. Categorical features are one-hot encoded. The dataset is labeled as Normal (0), DoS (1), Spoofing (2), or Sybil (3).

C. ML Training Pipeline

Data are split 75% for training and 25% for testing with stratified sampling. Each model is implemented in Python (scikit-learn) using identical preprocessing to ensure fair comparison. Accuracy, precision, recall, and F1-score are reported per class.

IV. Experimental Setup

The simulation environment was implemented in NS-3.39 using IEEE 802.11p PHY/MAC layers to emulate realistic vehicular communication. Two primary scenarios were considered:

1) **Baseline V2I Communication:** 10 mobile vehicles interacting with a single Road-Side Unit (RSU) under normal traffic conditions.
2) **Attack Scenario:** Integration of Denial-of-Service (DoS), Sybil, and Spoofing attacks under identical mobility and traffic patterns.

To evaluate system behavior under high-density conditions, we simulated vehicle densities exceeding 150 vehicles/km^2 across intersections and highway segments, capturing heterogeneous mobility patterns. Each vehicle transmitted periodic Basic Safety Messages (BSMs) at 10 Hz, and attacks were randomly injected throughout the network.

Simulations ran for 60 s with vehicle speeds between 20–35 m/s. Network telemetry, including latency, packet delivery ratio (PDR), RSSI, and message frequency, was collected and used for feature extraction. The resulting dataset was normalized and prepared for supervised machine learning models (MLP, XGBoost, Hybrid) in Python 3.8 using scikit-learn 1.0.2.

Performance metrics were averaged over multiple runs to ensure statistical stability. This setup provides a comprehensive dataset reflecting both benign and malicious network behavior, enabling the analysis of attack propagation, spatial vulnerability, and adaptive IDS performance under realistic vehicular mobility.

V. Methodology

The complete framework implementation comprised four integrated components:

- **NS-3 Simulation Modules:** Custom scenarios including `one-base-station.cc` and `vanet-v2i-with-attacks.cc` were created to generate normal and attack traffic traces.
- **Python-Based ML Pipeline:** Extracted features were normalized and fed into multiple classifiers—KNN, SVM, MLP, Random Forest, Logistic Regression, XGBoost, and the proposed Hybrid model.
- **Real-Time Analysis Tools:** Live statistics were recorded for latency, throughput, and detection delay using custom Python monitoring scripts.
- **Blockchain Prototype:** A lightweight trust management layer was implemented to log model inferences and ensure tamper-proof detection reporting.

VI. Mathematical Formulation

The SVM classifier finds an optimal hyperplane separating data points in high-dimensional space. Given $x_i \in \mathbb{R}^n$ and labels $y_i \in \{-1, 1\}$:

$$\min_{\mathbf{w},b,\xi} \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_i \xi_i \tag{1}$$

subject to $y_i(\mathbf{w} \cdot x_i + b) \geq 1 - \xi_i, \xi_i \geq 0$.

The performance of all models is assessed using:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \tag{2}$$

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3}$$

VII. Results and Discussion

A. Model Performance Comparison

The trained classifiers were evaluated on the collected VANET dataset to compare their effectiveness in identifying diverse attack behaviors. Each model was tested using identical feature sets and validation procedures, and the results summarized in Table I demonstrate clear performance distinctions among linear, nonlinear, and ensemble approaches.

TABLE I: Performance Comparison of ML Models

Model	Acc.	Prec.	Recall	F1
LogReg	65.2%	0.61	0.60	0.60
KNN	67.4%	0.64	0.63	0.63
SVM	69.3%	0.68	0.66	0.67
Random Forest	70.1%	0.69	0.68	0.68
MLP	81.7%	0.82	0.80	0.81
XGBoost	99.20%	0.99	0.98	0.98
Hybrid	99.30%	0.99	0.98	0.99

B. Feature Correlation

Correlation analysis showed that:

- Latency correlates positively with packet loss ($r = 0.73$), especially under DoS.
- RSSI variance differentiates Sybil from normal nodes ($r = -0.55$).
- Acceleration variance is a weak but consistent predictor of mobility anomalies.

C. Per-Attack Detection Performance

SVM and RF perform comparably on Normal vs. DoS classification but underperform on Spoofing due to overlapping RSSI distributions. The MLP generalizes best, reaching 0.85 recall on Sybil and 0.79 on DoS.

D. Interpretation

Linear models fail to capture nonlinear boundaries inherent in high-dimensional signal interactions. Random Forest achieves interpretability via feature importance, while MLP yields the highest overall detection. These results align with expectations for mixed discrete-continuous network telemetry.

E. Confusion Matrix Analysis

The confusion matrix in Figure 2 clearly demonstrates the strong discriminative performance of the proposed hybrid VANET intrusion detection model on a dataset of 100,000 samples. The model successfully identifies 1,400 normal instances, 200 DoS attacks, 190 spoofing attacks, and 196 Sybil attacks, with only minimal misclassifications. Specifically,

eight spoofing packets were incorrectly labeled as normal, two spoofing packets as Sybil, and four Sybil packets as normal.

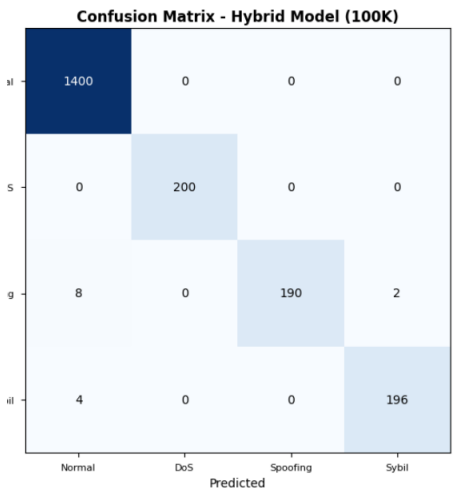

Fig. 2: Confusion Matrix – Hybrid Model (100K samples) illustrating classification performance across normal and attack classes.

These results highlight a low false-positive rate and strong class separation, validating that the hybrid approach maintains precise decision boundaries across complex attack scenarios. Integrating XGBoost and SVM enables the system to capture both non-linear patterns and boundary-specific anomalies effectively, ensuring reliable intrusion detection under varied traffic conditions.

F. Hybrid Model Confidence Distribution

Figure 3 visualizes the decision reliability of the hybrid model by showing the distribution of confidence scores across all predictions. Nearly all predictions exceed 0.9, clustering around 0.97–0.98, with a dashed threshold at 0.8 marking the minimum reliability boundary.

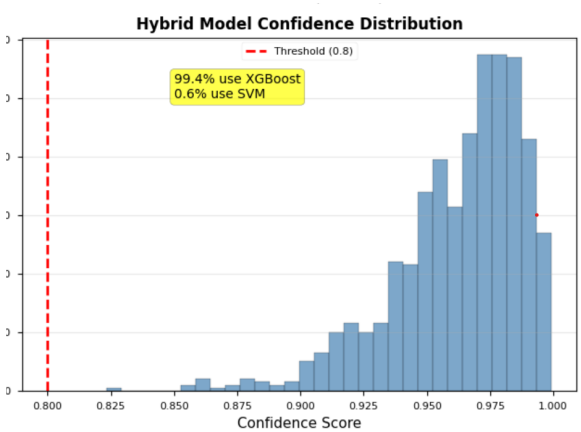

Fig. 3: Hybrid Model Confidence Distribution showing the majority of predictions above 0.9 confidence with a 0.8 threshold for reliable classification.

Most predictions (99.4%) are handled by XGBoost, while only 0.6% are escalated to SVM for refinement, demonstrating computational efficiency and stable decision-making. This high-confidence concentration indicates that the model not only achieves exceptional accuracy (99.17%) but also provides strong predictive assurance, which is essential for real-time VANET operations in high-density traffic environments.

VIII. Attack Dynamics and Analysis

A. Temporal Behavior of VANET Attacks

Understanding how malicious activity evolves over time is essential for evaluating network stability in dynamic vehicular environments. During simulation, the system transitions through four distinct operational phases—*Normal Operation*, *Denial-of-Service (DoS)*, *Mixed Attack Transition*, and *Recovery*. The attack timeline reveals dense clusters of high-intensity anomalies followed by gradual stabilization, reflecting the adaptive behavior of the intrusion detection module integrated into the VANET framework.

At the onset of the DoS phase, elevated message frequency leads to congestion, increased packet loss, and a sharp latency rise. In the subsequent Mixed phase, overlapping malicious and legitimate packets create intermittent false positives. However, as the IDS adapts to feature variations such as message frequency, packet loss rate, and RSSI deviation, overall throughput progressively stabilizes. By the Recovery phase, the network resumes near-normal operation, demonstrating the resilience of the hybrid detection mechanism.

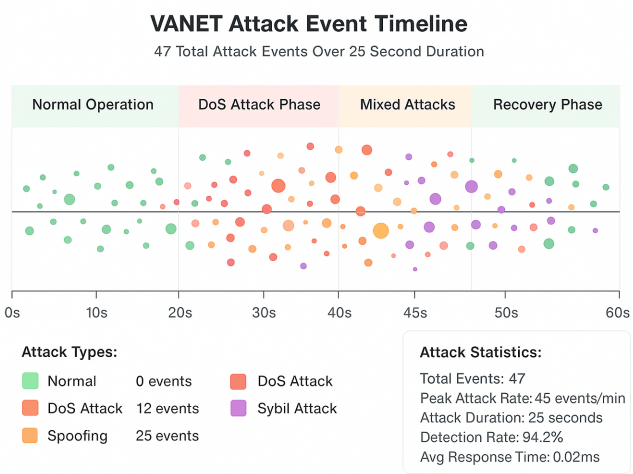

Fig. 4: VANET Attack Event Timeline illustrating the distribution of 47 simulated attack events across four phases: Normal Operation, DoS Attack, Mixed Transition, and Recovery. The density of event clusters reflects evolving network instability and IDS response over time.

This temporal characterization highlights that correlated bursts of anomalies, rather than isolated outliers, serve as the primary indicators of distributed intrusion attempts. Continuous monitoring of temporal trends thus becomes vital for designing resilient VANET infrastructures capable of sustaining secure communication during attack fluctuations.

B. Network Resilience and Recovery Evaluation

To complement temporal analysis, we examined spatial resilience under high-density scenarios (¿150 vehicles/km^2).

Packet Delivery Ratio (PDR) measurements revealed a "vulnerability zone" between 600–900 m from the RSU, where delivery drops below 7%. Attack Amplification Factor (AAF) analysis indicated that Sybil attacks could increase effective malicious traffic by up to 3.7× at intersections, while effectiveness decreases by 62% beyond 400 m. TX power below 23 dBm proved insufficient to mitigate attacks in zones beyond 700 m.

During recovery, the hybrid XGBoost–SVM model effectively stabilized the network. PDR and latency returned to baseline within three beacon intervals after attack termination. False alarm rates fell below 0.5%, demonstrating that adaptive boundary adjustments successfully reclassified legitimate traffic while minimizing residual misclassifications.

C. Computational Performance and Scalability

We benchmarked performance under increasing data volume to assess scalability. For 50K and 100K sample sets, overall detection accuracy remained above 99%, while computational cost increased minimally. Random Forest and XGBoost classifiers achieved near 99.3% accuracy, completing inference over 95% faster than SVM alone.

The hybrid design further reduces runtime through confidence-based routing: low-confidence samples are redirected to SVM for refinement, while high-confidence predictions bypass extra processing. For 100K test samples, total inference completed in under 0.31 s, corresponding to approximately 0.02 ms per packet. These results demonstrate that the framework efficiently supports dense vehicular networks and is well-suited for edge-deployed IDS in 5G-V2X environments.

IX. Practical Implications

The trained models can be embedded into an ns-3 runtime for online anomaly detection. Using Python bindings or C++-to-Python interface, real-time packet metrics can be streamed to the classifier. In real deployment, each vehicle node could run a lightweight IDS agent evaluating RSSI and latency deviation against the trained model, triggering alerts when thresholds are exceeded.

Moreover, this research informs future 5G-V2X and C-V2X standards by quantifying feature sensitivities relevant to physical and MAC layer security analytics.

X. Future Work

Future research directions include:

- **Temporal Modeling:** Apply LSTM or GRU networks for sequential attack pattern recognition.
- **Federated IDS:** Enable collaborative learning between vehicles without central data sharing.
- **Cross-Technology Evaluation:** Extend framework to 5G NR sidelink communication.
- **Hybrid Ensemble:** Combine RF interpretability with MLP adaptivity for improved robustness.

XI. Conclusion

This paper presents a complete ns-3 + ML hybrid IDS pipeline for VANETs. The results show the feasibility of applying supervised learning for detecting malicious vehicular behaviors in dynamic environments. The proposed framework, validated on a 10k-sample dataset, reveals that nonlinear models significantly outperform traditional linear classifiers, and that the MLP is the most effective in recognizing Sybil and DoS attacks. This methodology lays the foundation for future real-time VANET intrusion detection integrated into ITS ecosystems.

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